CHAPTER 3

Innocent (Commutative) Algebra

"The introduction of the cipher 0 or the group concept was general nonsense too, and mathematics was more or less stagnating for thousands of years because nobody was around to take such childish steps..."
— Alexander Grothendieck (in his 1982 letter to Ronald Brown)

"Taking a new step, uttering a new word, is what people fear most."
— Fyodor Dostoevsky, Crime and Punishment

3.1. It will not be possible for us to discuss the complete (basic) algebra. We will assume some knowledge and this chapter should be treated to recall some of the basic facts. If one is not comfortable with algebra of rings, fields, vector spaces (and modules), then one may refer to [**1**, **2**]. Even though the chapter is set for algebra, we take some irregular detours to other parts of mathematics.

1. Some Preliminaries

Definition 3.2 (Subset). For two sets S and T we say that S is a *subset* of T if each element of S is also an element of T, i.e. if S $\subseteq$ T $\forall$ $x \in S$ we have $x \in T$.
Simultaneously, we can also say that T contains S which can be written as $T \supseteq S$. If $S \subseteq T$ and $S \neq T$ hold, then we write $S \subset T$ and we say that S is a *proper subset* of T.

Remark 3.3. Two sets are equal iff each is a subset of the other. In symbolic notation: $(A = B) \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$

Definition 3.4 (Cartesian Product). Consider a given family of sets $(S_i)_{i \in I)}$, where I denotes the index set, then the cartesian product denoted as $\prod_i S_i$, of the family of given sets is the set of all functions f from the index set I with f_j in each S_j, for each $j \in I$.

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For a finite family of sets, we can define the Cartesian Product as:

$$\prod_{i=1}^{i=n} S_i = \{(s_1, s_2, s_3, \cdots, s_n) \mid s_i \in S_i,\ \forall i \in \{1, 2, 3, \cdots, n\}\}$$

DEFINITION 3.5 (Relation). A relation is a subset of the cartesian product of two sets that helps to establish a connection between the elements of the sets. Given a set X, a relation R over X is a set of ordered pairs of elements from X:

$$\mathbb{R} \subseteq \{(x, y) \mid x, y \in X\} \tag{1}$$

DEFINITION 3.6 (Injection, Surjection and Bijection). A function $f : S \hookrightarrow T$ is *injective* or one-to-one if no element of T (no second coordinate) appears in more than one ordered pair. Such a function is called an **injection**. A function $f : S \twoheadrightarrow T$ is *surjective* or onto if every element of T appears in an ordered pair. Such a function is called a **surjection**.

A function that is both surjective and injective is said to be *bijective*. Bijective functions are called **bijections**.

3.7. One should be careful that all of these definitions are apt for set theoretic foundations only. We would revise most of these definitions in future sections, especially in category theory.

DEFINITION 3.8 (Equivalence Relation). A binary relation[1] $\sim$ on a set is said to be an equivalence relation, iff it is *reflexive, symmetric* and *transitive*. An equivalence class of an element a is defined as:

$$[a] = \{x \in X :\ a \sim x\}. \tag{2}$$

The set of all *equivalence classes* in X with respect to an equivalence relation R is denoted as X/R and is called X modulo R.

3.9. Let us write the axioms[2] for reflexive, irreflexive, symmetric, anti-symmetric, asymmetry and transitive relations.

Reflexive: $\forall a,\ a \sim a$
Symmetric: $\forall a, b,\ a \sim b \implies b \sim a$
Transitive: $\forall a, b, c,\ a \sim b$ and $b \sim c \implies a \sim c$
Asymmetry: $\forall a, b,\ a \sim b \implies \neg\ b \sim a$
Anti-symmetry: $\forall a, b,\ a \sim b,\ b \sim a \implies a = b$
Irreflexive: $\forall a,\ \neg a \sim a$ (for example $(\mathbb{R}, <)$)

[1]In formal logic, a *predicate* or relation is an n-ary symbol and so is an operation. Binary operation is just an example of n-ary operations.

[2]Given any relation, there are some properties that we list down which are called axioms. For instance, axioms of $(\mathbb{R}, \leq)$ consists of reflexivity, anti-symmetry, and transitivity.

REMARK 3.10. The symbol '$\neg$' denotes the negation statement.

DEFINITION 3.11 (Congruence Relation). If a and b have the same remainder when they are divided by a non-zero integer m, then we say that *a is congruent to b mod m*, represented as:

$$a \equiv b(mod\ m) \tag{3}$$

Congruence modulo m is an equivalence relation which is compatible with the operations of addition, subtraction, and multiplication.

DEFINITION 3.12 (Residue Class). The residue class of a mod m is denoted as $[a]$, where $[a]$ represents the set of all integers that are congruent to a modulo m.

DEFINITION 3.13 (Partial Order and Total Order). The reflexive, antisymmetric and transitive relation $\mathcal{R}$ is called a partial order relation and $\mathcal{R}$ is said to define a partial ordering of $\mathcal{S}$. The addition of the trichotomy law to the set of axioms of the partial order, defines the total order.

Hence, for a set $\mathcal{S}$ to be totally(/partially) ordered, it must have a relation $\mathcal{R}$ on it with the total(/partial) order.

EXAMPLE 3.14. $(\mathbb{R}, <)$ is an example of strict linear order which is totally ordered and $(\mathbb{R}, \leq)$ is an example of linear order which is partially ordered.

LEMMA 3.15 (Kuratowski-Zorn Lemma a.k.a. Zorn's Lemma). If every totally ordered subset of a poset $\mathcal{S}$ has an upper bound, then $\mathcal{S}$ contains a maximal element.

2. Algebraic Structures

DEFINITION 3.16 (Algebraic Structure). An algebraic structure is defined as a non-empty set that forms the closure with at least one operation, denoted as $(S, \star)$, where S denotes the non-empty set and $\star$ defines the operation.

In accordance with the above definition, any non-empty set with an operation will form an algebraic structure and with the duality principle, will also possess a geometrical interpretation.

DEFINITION 3.17 (Semigroup, Monoid, Groupoid). A *semigroup* is an algebraic structure equipped with a binary operation and satisfying associativity.

A *monoid* is a semigroup fulfilling the existence of a unique identity.

A *groupoid* is a monoid consisting of an inverse map with respect to every map that exists.

3.18. A group can be seen as a set of *symmetries* of any object. And symmetries, on the other hand, will be defined as mapping an entity to itself, while preserving the structure.

Definition 3.19 (Group). A group is a set G with a binary operation $G \times G \Rightarrow G$, usually written as

$$(a, b) \Rightarrow a + b, a \times b, ab, \text{ or } a \cdot b, \tag{4}$$

and must follow the following *axioms*:

(1) $\exists$ a unique identity $(e, 1, \text{ or } 0)$ such that $ea = a = ae$
(2) $\exists$ a 2-sided inverse, for every element a^{-1} such that $a^{-1}a = aa^{-1} = e$
(3) $\exists$ associativity amongst the elements such that $(ab)c = a(bc)$

3.20. Alternatively, we can also see a group as a monoid, consisting of a 2-sided inverse of all the elements in the underlying set.

Definition 3.21 (Commutative Group). Let $(G, \star)$ be a group, then G is said to be a commutative group if $a \star b = b \star a, \ \forall\ a, b \in G$.

3.22. We say that a group structure $(G, \star)$ is **Abelian** if it is commutative. That is, in an Abelian Group, the order of applying the group operation on the elements becomes irrelevant. Hence, the elements have become invariant under the order of application of the group operation.

Example 3.23 (Example of Finite Abelian Group). A group G defined as $(G = \{e, a, b, c\}, \star)$ [3], where $\star : G \times G \longrightarrow G$, such that each element is self inverse and when two different elements other than identity are multiplied, then the third element, other than the identity is the result.

Definition 3.24 (Order of an Element). For a group G and $x \in G$, the order of x is defined as the smallest positive integer n such that $x^n = e$, where e is the identity of G. It is denoted as $o(x)$.

If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order.

3.25. An element of a group has order 1 iff it is the identity.

Definition 3.26 (Order of a Group). For a discrete group G, the **order** of the group is the cardinality of the underlying set. Hence, for a finite group G, its order $|G|$ is a natural number.

3.27. The *order of a subgroup* is similar to a group, i.e., it represents the number of the elements present in the subgroup.

Theorem 3.28 (Fundamental theorem of order). Let G be a group and $g \in G$ be an element then,

[3] Klein four-group

(1) If $|g| = \infty$, then $g^n = g^m$ iff $n = m$.
(2) If $|g| = k$ is finite, then $g^n = g^m$ iff $k|(n - m)$.

Definition 3.29 (Group Generator). A group generator is a subset of a group, such that every element of the group can be expressed as a finite combination of elements and their inverses. This can be expressed notationally as:

$$G = \langle S \rangle \tag{5}$$

and we say that *G is generated by S* or *S generates G*.

Definition 3.30 (Relations). Any equations in a general group G that the generators satisfy are called **relations** in G.

Example 3.31. For the group $\mathbb{Z}$, the set of all integers, both -1 and 1 are generators, as every integer number can be reached by repeatedly adding or subtracting these values.
Similarly, the cyclic group Z_n consists of integers modulo n, and a single generator can generate it, namely 1 or -1.

Example 3.32. In the complex numbers group $\mathcal{C}$, under multiplication, the unit complex numbers $e^{i\theta}$ where θ is a real number, can generate all the other elements.
Since, any complex number can be written in the polar form $re^{i\theta}$, where r is a real number representing the magnitude and θ is an argument.

Definition 3.33 (Subgroup). Consider a group G, and a subset H of G. H is a subgroup of g, iff H also froms a group under the same operations as in G.

3.34. S generates G iff the smallest subgroup of G containing S is G itself (the improper subgroup).

- A *proper subgroup* is the subgroup H which is the proper subset of G, i.e., $G \neq H$, represented as, $H < G$.
- Since $G \subseteq G \Rightarrow (H = G, \star)$ is also a subgroup of G, referred to as the *improper subgroup* of $(G, \star)$.
- A *trivial subgroup* is the subgroup $\{e\}$, consisting of just the identity element.
- If H is a subgroup, then G would be referred to as the *overgroup* of H.

Definition 3.35 (Relations). Any equations in a general group G that the generators satisfy are called relations in G.

3.36. To check if a set H equipped with a binary operation forms a subgroup of a group G, one needs to check for the presence of the identity elements, closure under the law of composition and for the existence of inverses.

Definition 3.37 (Normalizer). Given a subset S of the underlying set of a group G, its normalizer consists of all the elements from G such that the left and the right action by those elements on S, results in the same subsets.

$$N(S) = N_G(S) = \{n \in G : Sn = nS\}$$

A normalizer forms a subgroup of G.

DEFINITION 3.38 (Centralizer). Consider a group G, and a subset S of G, i.e., a subset of it's underlying set, then the centralizer subgroup of S in G is the subgroup of all the elements c in G which commute with all the elements of S.

$$C_G(S) = \{c \in G : cs = sc\}$$

THEOREM 3.39. Given a subset $S \subset G$ of a group G, then the centralizer subgroup of S is a subgroup of the normalizer subgroup such that,

$$C_G(S) \subset N_G(S)$$

DEFINITION 3.40 (Stabilizer). Consider a group G an let us define an action by the group G on a set X for every element $x \in X$ such that,

$$G \times X \to X$$

$$Stab_G(x) = \{g \in G : g \cdot x = x\} \tag{6}$$

Hence, a stabilizer subgroup of X consists of a set of elements under which all the elements of X are conjugation invariant.

3.41. Now, a stabilizer subgroup, also known as the **isotropy subgroup** of x is defined as the set of all the elements in G that leave x fixed. In other words, the action *stabilizes* every element of X, in other words.

Now, if all the stabilizer groups are trivial, then the action is called a **free action**.

3.42. Isotropy refers to a sense of uniformity in all orientations. In mathematical terms, it (G_x) is defined as exactly the above equation(6). Moreover, when two points x and y are on the same group orbit, say $y = gx$. then the isotropy groups are conjugate subgroups, that is, $G_y = gG_xg^{-1}$. More precisely, any subgroups conjugate to G_y occurs as an isotropy group G_y to some point y lying on the same orbit as x.

THEOREM 3.43. Let $a \in G$, and $(G, \cdot)$ be a finite group, then the set generated by the power of a defined as: $S = \{a^n : n \in \mathbb{Z}, a \in G\}$ is finite.

PROOF. Let $a \in G$. Then, $\cdots, a^{-3}, a^{-2}, a^{-1}, a^0, a^1, a^2, \cdots \in G$ where, $n \in \mathbb{Z}$. By closure property, the above holds. Now, $(G, \cdot)$ is a finite group, hence all the elements are not distinct. Let, $r < s : a^r = a^s$, then left action by a^{-r} gives us:

$$a^{-r} \cdot a^r = a^{-r} \cdot a^s \tag{7}$$

$$e = a^{-r} \cdot a^s = a^{s-r} \tag{8}$$

$$\therefore \; o(a) \leq (s - r) \tag{9}$$

The integral powers of a will result into non-distinct values. Hence, S is a finite set. $\square$

3.44. Let G, G' be monoids (sets with a law of composition which follows associativity and has an identity), then a **monoid-homomorphism** of G into G' is defined as a map $f : G \to G'$, such that $f(xy) = f(x)f(y)$ for all $x, y \in G$ and the identity in G is mapped to the identity in G'

3.45. If G and G' are groups, then the **group homomorphsim** between G and G' is simply the monoid homomorphism.

3.46. Let G and G' be groups and let $f : G \to G$" be a group homomorphism, then for all $x, y \in G$,

(1) $f(xy) = f(x)f(y)$
(2) $f(x^{-1}) = f(x)^{-1}$
(3) $e' = f(e) = f(xx^{-1}) = f(x)f(x^{-1}) = f(x)f(x)^{-1}$, where e' is the identity in G' and e is the identity in G.

The composition of two group homorphisms f and g, defined as $f \circ g$ or $g \circ f$ (if compatible,) is also a group homomrophism.

Note that for a cyclic group G, if $G =< x >$, then $G' =< f(x) > \Leftrightarrow f : G \to G'$ is a surjective homomorphism.

3.47. Let G, G' be groups, then a homomorphism $f : G \to G'$ is called an **isomorphism** if there exists a group homomorphism $g : G' \to G$ such that $f \circ g = 1_{G'}$ and $g \circ f = 1_G$, where 1_G and $1_{G'}$ represent **identity morphisms** on G and G', respectively.

Note that, it is not enough to say that the morphism is invertible for an isomorphism to exist, we need to ensure that the composition of the two morphisms gives the identity on the domain of the composition map.

For $f \circ g = 1_{G'}$, we need g to be an injective morphism and f to be a surjective morphism. Also, f is an isomorphism $\Leftrightarrow f$ is bijective.

An isomorphism ensures that two groups are same up to re-naming/representation of their elements. That is to say, under the isomorphism f, G and G' become identical, and the identity in G is mapped to the identity in G' and the order of the respective elements is also preserved.

3.48. If for a given isomorphism $f : G \to G'$, if $G = G'$, then f is said to be an **automorphism**, i.e., an isomorphism from the group G to itself.

Similarly, a homomorphism from the group G to itself is called an **endomorphism**.

Example 3.49. Consider the additive group $\mathbb{Z}/5\mathbb{Z} = \{0, 1, 2, 3, 4\}$ with the binary operation addition modulo 5. Let us define an endomorphism $f : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$ such that

$f(x) = x^x$. Then $f(0) = 0,\ f(1) = 1,\ f(2) = 4,\ f(3) = 2,\ f(4) = 1$. Here, $x \mapsto x^x$ is referred to as the $x-$**th power map**.

EXAMPLE 3.50. Let A be an abelian group. Define a map $f : A \to A$ such that $a \mapsto a^{-1}$. Then, clearly f is an endomorphism. And,

$$f(a_1a_2) = (a_1a_2)^{-1} = a_2^{-1}a_1^{-1} = a_1{-}1a_2^{-1} = f(a_1)f(a_2) = f(a_2)f(a_1) = f(a_2a_1)$$

is therefore a homomorphism. See $f(f(a)) = f(a^{-1}) = (a^{-1})^{-1} = a$. Therefore,

$$f \circ f = \mathrm{id}_A$$

and hence, f is injective and surjective. Moreover, $f^2 = \mathrm{id}$ also implies that f is an involution[4]. f is therefore an **automorphism**.

3.51. Consider the map $f : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$, such that $x \mapsto x(\mathrm{mod}\ 2)$. In this case, we can collect all those elements of $\mathbb{Z}$ which map to $0 \in \mathbb{Z}/2\mathbb{Z}$, the set of all such elements under the map f is referred to as the **kernel** of f, denoted $\ker f$. Similarly, we can create a set of all those elements of $\mathbb{Z}$ which *do not* map to $0 \in \mathbb{Z}/2\mathbb{Z}$. We call this the **image** of f, denoted $\mathrm{im}\ f$.

DEFINITION 3.52. (Kernel and Image). Let $f : G \to H$ be a group homomorphism. The **kernel** of f is defined as,

$$\ker f := \{a \in G \mid f(a) = e \in H\}$$

image of f is defined as,

$$\mathrm{im} f := \{b \in H \mid b = f(a)\ \forall a \in G\}$$

.

3.53. Let us define a group homomorphism $f : \mathbb{Z} \to 2\mathbb{Z}$, such that $x \mapsto 2x$. Then, only $0 \in \ker f$. In such a case, we say that the kernel is **trivial**. Clearly, f is also an injective morphism.

Whenever the kernel of a homomorphism is trivial, the map is injective.

EXAMPLE 3.54. Define a group homomorphism $f : G \to H$ such that $\ker f = \{e\}$, where e is the identity in G.

Consider $x, y \in G$ and let $f(x) = f(y)$, and multiply by $f(x^{-1})$ on the left:

$$f(x^{-1})f(x) = f(x^{-1})f(y) = f(x^{-1}y) = f(e) = e'$$

where e' is the identity in H. Hence, $x^{-1}y \in \ker f$, and $x^{-1}y = e$. Thus, $x = y$ and the map is injective.

3.55. If in 3.54, the f is also surjective, then f is an isomorphism.

[4]An involution is a function which is it's own self-inverse.

3.56. A homomorphism $f : G \to G'$ which establishes an isomorphism between G and its image in G', is referred to as an **embedding**. Thus, an injective homomorphism is an embedding, denoted as

$$f : G \hookrightarrow G'.$$

DEFINITION 3.57. (Inverse Image.)

Let $f : G \to H$ be a group homomorphism. If B is a subset of H, then the inverse image of B under f is determined as

$$f^{-1}(B) := \{a \in G \mid f(a) \in B\}$$

EXAMPLE 3.58. Let $f : G \to H$ be a group homomorphims. Then ker$f$ is a subset of G. Let $g_1, g_2 \in$ ker$f$. Then,$f(g_1) = f(g_2) = e'$ in H. Then, for $g_1 g_2 = g_3$, we have

$$f(g_3) = f(g_1 g_2) = f(g_1)f(g_2) = e'$$

and hence ker$f$ satisfies closure property. And we know $f(e) = e'$.

$$\Rightarrow f(g_1 g_1^{-1}) = e' = f(g_1)f(g_1^{-1}) = e' f(g_1^{-1}) = f(g_1^{-1}) = f(g_1)^{-1}$$

Therefore, ker$f$ also consists of inverses. Hence, ker$f$ *forms a subgroup of* G.

DEFINITION 3.59. (Cosets.) Let G be a group and H a subgroup. A **left coset** of H in G is a subset of G, denoted as aH for some $a \in G$ such that

$$aH := \{ah, \ \forall\ h \in H\}.$$

Dually, a **right coset** of H in g is a subset of G, denoted as Hb for some $b \in G$ such that

$$Hb := \{hb, \ \forall\ b \in G\}.$$

3.60. Let H be a subgroup of a group G, and let aH be a left coset of H in G. Then the morphism $f : H \to aH$ such that $h \mapsto ah$ induces a bijection of H onto aH. Hence, any two left cosets (or right cosets) have the same cardinality.

EXAMPLE 3.61. Let G be a group and let H be a subgroup of G and let aH and bH be two left cosets of H in G. Let aH and bH have one element in common then, the two cosets are the same.

Let $ah_1 = bh_2$, then $a = bh_2 h_1^{-1}$

$$\Rightarrow aH = b(h_2 h_1^{-1})H = bH$$

since for any $x \in H$, we have $xH = H$.

3.62. If G is a group, then it is the disjoint union of the left cosets of H, where $H < G$. Similarly, G can also be expressed as the disjoint union of the right cosets of H.

DEFINITION 3.63. (Index.) The number of left or right cosets of a subgroup H in a group G is expressed as $(G : H)$ and is called the (left/ right) **index** of H in G.

3.64. By the definition of index above, the index of a trivial subgroup of a group G is $(G:1)$ which is precisely the order of the group G.

EXAMPLE 3.65. Let $G = \mathbb{Z}$ be the group of integers under addition, and let $H = 2\mathbb{Z}$ be the subgroup consisting of even integers. Then $2\mathbb{Z}$ has two cosets in $\mathbb{Z}$ namely the set of even integers and the set of odd integers, so the index $|\mathbb{Z} : 2\mathbb{Z}|$ is 2.
More generally,

$$|\mathbb{Z} : n\mathbb{Z}| = n$$

for any positive integer n.

THEOREM 3.66 (Lagrange's Theorem). *Let G be a finite group and let $H \leq G$. Then $|H|$ divides $|G|$ and, more precisely, $|G| = [G : H]|H|$.*

PROOF. For each $g \in G$, consider the left coset $gH = \{gh : h \in H\}$.

Define

$$\varphi_g : H \longrightarrow gH, \qquad \varphi_g(h) = gh. \tag{10}$$

This map is surjective by the definition of gH.

To show injectivity, suppose

$$\varphi_g(h_1) = \varphi_g(h_2).$$

Then

$$gh_1 = gh_2.$$

Multiplying on the left by g^{-1} gives

$$h_1 = h_2.$$

Hence φ_g is bijective, and therefore

$$|gH| = |H|.$$

Furthermore, any two left cosets of H are either equal or disjoint. Thus the left cosets of H form a partition of G:

$$G = \bigcup_{gH \in G/H} gH.$$

Since there are $[G : H]$ left cosets and every coset contains exactly $|H|$ elements, we obtain

$$\begin{aligned} |G| &= \sum_{gH \in G/H} |gH| \\ &= \sum_{gH \in G/H} |H| \\ &= [G : H]|H|. \end{aligned}$$

Therefore, $|H| \mid |G|$. $\square$

3.67. As a consequence of Lagrange's theorem, we obtain the following corollary: Let G be a finite group and let $g \in G$. Then $|g| \mid |G|$.

The idea of the proof is to start wiht a cyclic subgroup of the gorup G generated by a single element.

Definition 3.68. (Normal Subgroup.) A subgroup H of a group G is called **normal** if every element of G *normalises* H, i.e., if $gHg^{-1} = H$ for all $g \in G$.

If H is a normal subgroup of G we denote it as $H \triangleleft G$.

3.69. Let $f : G \to H$ be a group homomorphism. Then the kernel of f is a normal subgroup of G.

3.70. If for each $g \in G$ and each $B \subset G$ let $g : B \mapsto gBg^{-1}$ where $gbg^{-1} = \{gbg^{-1} | b \in B\}$, then G is said to act on B by **conjugation**.

Definition 3.71. (Group Action.) A **(left) group action** of a group G on a set A is a map $G \times A \to A$, for all $g \in G$ and $a \in A$, such that the following hold.

(1) $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$
(2) $e \cdot a = a$

Dually, **(right) group action** of a group G on a set A is a map $A \times G \to A$, for all $g \in G$ and $a \in A$, such that the above hold (respectively).

Example 3.72. Consider the left group action defined as, $\sigma : G \times A \to A$ such that $\sigma_g(a) = g \cdot a$. Then, for all $a \in A$,

$$\begin{aligned}(\sigma_{g^{-1}} \circ \sigma_g)(a) &= \sigma_{g^{-1}}(\sigma_g(a)) \\ &= g^{-1} \cdot (g \cdot (a)) \\ &= (g^{-1} g) \cdot a \\ &= e \cdot a = a\end{aligned} \tag{11}$$

Hence, $(\sigma_{g^{-1}} \circ \sigma_g)$ is the identity map. Since g was arbitrary, $(\sigma_g \circ \sigma_{g^{-1}})$ is also the identity map. σ_g has a two-sided inverse.

Essentially, σ_g is a permutation of A.

Let us consider S_A to be the set of all the possible permutations of the elements of A. Define $\psi : G \to S_A$ such that $\psi(g) = \sigma_g$.

This map is well-defined, as $\sigma_g \in S_A$. Consider

$$\begin{aligned}\psi(g_1g_2)(a) &= \sigma_{g_1g_2}(a)\\ &= (g_1g_2)\cdot(a)\\ &= g_1\cdot(g_2\cdot a)\\ &= \sigma_{g_1}(\sigma_{g_2}(a))\\ &= (\psi(g_1)\circ\psi(g_2))(a)\end{aligned} \tag{12}$$

Therefore, ψ is a homomorphism.

3.73. The homomorphism discussed in 3.72 is called the **permutation representation** associated to the specified action.

Example 3.74. Let $ga = a$ for all $g \in G$, $a \in A$. This action is called the **trivial action** of the group G on A, and the associated permutation representation $G \to S_A$ is the *trivial* homomorphism, mapping every element of A to itself.

3.75. If for each $g \in G$ and each $B \subset G$ let $g : B \mapsto gBg^{-1}$ where $gbg^{-1} = \{gbg^{-1} | b \in B\}$, then G is said to act on B by **conjugation**.

3.76. If G acts on a set B non-trivially, such that different elements of G induce a distinct permutation of B, then G acts **faithfully** on B. Here, the associated permutation representation is *injective.*

Definition 3.77. (Kernel of a group action.) For the group action $G \times B \to B$, we can define the **kernel** to be

$$\{g \in G \mid gb = b, \ \forall\ b \in B\}$$

We say that the kernel consists of all the elements of G which *fix* all the elements of B.

3.78. For the trivial action, the kernel is the group G, and such an action is by definition, **unfaithful**.

That is, whenever $|\ker| > 1$, the group action is unfaithful.

3.79. By definition, a normal subgroup is conjugation invariant. Hence for a group G and it's normal subgroup H, we must have $xHx^{-1} = H$ for all $x \in G$.

Example 3.80. Let $f : G \to G'$ be a homomorphism and let $\ker f$ be it's kernel. From (**??**) we know that $\ker f < G$. Then for $x, y \in G$

$$x(\ker f) := \{xa \in G \mid f(a) = e' \in G'\}$$

and

$$(\ker f)y := \{ay \in G \mid f(a) = e' \in G'\}$$

Hence,

$$f(xa) = f(x)f(a) \\ = f(x)e' = f(x) \tag{13}$$

and

$$f(ay) = f(a)f(y) \\ = e'f(y) = f(y) \tag{14}$$

Therefore, $x(\ker f) = (\ker f)x \;\Rightarrow \ker f \lhd G$.

3.81. Given a homomorphism $f : G \to H$, $\ker f$ forms a **normal subgroup** of G.

3.82. Before moving ahead with the concept of a quotient group, we will look into group action on a set. More specifically, we will understand by the following theorem, how a group acting ona set is quivalent to defining a homomorphism from the group to the symmetric group of the set.

THEOREM 3.83. *Let G be a group acting on a set X. Then there exists a group homomorphism*

$$\varphi : G \longrightarrow S_X,$$

where S_X denotes the group of all permutations of X.

PROOF. For each $g \in G$, define a map

$$\varphi_g : X \longrightarrow X \qquad \text{by} \qquad \varphi_g(x) = g \cdot x. \tag{15}$$

We first show that φ_g is a permutation of X. Since $g^{-1} \in G$, consider the map

$$\varphi_{g^{-1}}(x) = g^{-1} \cdot x.$$

For every $x \in X$,

$$\begin{aligned} \varphi_g(\varphi_{g^{-1}}(x)) &= g \cdot (g^{-1} \cdot x) \\ &= (gg^{-1}) \cdot x \\ &= e \cdot x \\ &= x. \end{aligned}$$

Similarly,

$$\varphi_{g^{-1}}(\varphi_g(x)) = x.$$

Hence $\varphi_{g^{-1}}$ is the inverse of φ_g, so $\varphi_g \in S_X$.

We may therefore define

$$\varphi : G \longrightarrow S_X, \qquad g \longmapsto \varphi_g.$$

It remains to show that φ is a homomorphism.

Let $g, h \in G$. For every $x \in X$,

$$\begin{aligned}\varphi(gh)(x) &= (gh) \cdot x \\ &= g \cdot (h \cdot x) \\ &= \varphi_g(\varphi_h(x)) \\ &= (\varphi_g \circ \varphi_h)(x).\end{aligned}$$

Therefore,

$$\varphi(gh) = \varphi_g \circ \varphi_h = \varphi(g)\varphi(h).$$

Thus φ is a group homomorphism.

Hence every group action gives rise to a homomorphism

$$\varphi : G \longrightarrow S_X.$$

□

DEFINITION 3.84 (Quotient Group). Let G be a group and N be a normal subgroup of G, then the group G/N is called the **quotient group** or the **factor group**, defined as

$$G/N := \{gN \mid g \in G\}$$

G/H is read as G *modulo* H, or G *mod* H.

3.85. A quotient group can be viewed as the group of all cosets of a normal subgroup.

THEOREM 3.86 (Orbit–Stabilizer Theorem). Let G be a group acting on a set X, and let $x \in X$. Then $|Gx| = [G : G_x]$, where $Gx = \{g \cdot x : g \in G\}$ is the orbit of x, and $G_x = \{g \in G : g \cdot x = x\}$ is the stabilizer of x.

PROOF. Define

$$\varphi : G/G_x \longrightarrow Gx, \qquad \varphi(gG_x) = g \cdot x. \tag{16}$$

We first show that φ is well-defined.

Suppose

$$gG_x = hG_x.$$

Then

$$h^{-1}g \in G_x,$$

so

$$(h^{-1}g) \cdot x = x.$$

Applying h gives

$$g \cdot x = h \cdot x.$$

The map φ is surjective by the definition of Gx.

If

$$\varphi(gG_x) = \varphi(hG_x),$$

then

$$g \cdot x = h \cdot x.$$

Hence

$$(h^{-1}g) \cdot x = x,$$

so $h^{-1}g \in G_x$, and therefore

$$gG_x = hG_x.$$

Thus φ is bijective, and consequently

$$|Gx| = |G/G_x| = [G : G_x].$$

□

Example 3.87. Let S_3 act on $\{1, 2, 3\}$ by permutation. Consider $x = 1$. Its orbit is

$$S_3 \cdot 1 = \{1, 2, 3\},$$

so

$$|S_3 \cdot 1| = 3.$$

The stabilizer of 1 is

$$(S_3)_1 = \{\sigma \in S_3 : \sigma(1) = 1\} = \{e, (23)\},$$

and hence

$$|(S_3)_1| = 2.$$

Since $|S_3| = 6$, we obtain

$$[S_3 : (S_3)_1] = \frac{6}{2} = 3.$$

Thus

$$|S_3 \cdot 1| = [S_3 : (S_3)_1] = 3,$$

as predicted by the Orbit–Stabilizer Theorem.

Definition 3.88 (Center). The centralizer of the group G is the center of the group G.

The **center** of G is the subgroup of G consisting of all elements of G which commute with all the other elements. The centre of a group is *normal*.

EXAMPLE 3.89. Let us consider the multiplicative group of all invertible $n \times n$ (square) matrices with entries from $\mathbb{R}$, called the **General Linear Group**, written as $\mathrm{GL_n}(\mathbb{R})$.

Then the determinant is a homomorphism from $\mathrm{GL_n}(\mathbb{R})$ into $\mathbb{R} - \{0\}$.

$$\psi : \mathrm{GL_n}(\mathbb{R}) \to \mathbb{R} - \{0\}$$

Since all the elements of $\mathrm{GL_n}(\mathbb{R})$ are invertible, the determinant of all the matrices will be non-zero. Moreover, since the multiplicative structure has been considered in $\mathbb{R}$, we have $1 \in \mathbb{R}$ to be the identity. The kernel of ψ would consist of all those matrices whose determinant is 1. In this case, $\ker\psi$ is the $\mathrm{SL_n}(\mathbb{R})$, which is the **special linear group**[5].

3.90. Let G be a finite group. Consider the conjugation action of G on itself, defined by $g \cdot x = gxg^{-1}, g, x \in G$.

For $x \in G$, its orbit is $Gx = \{gxg^{-1} : g \in G\}$, which is precisely the conjugacy class of x.

The stabilizer of x under this action is $G_x = \{g \in G : gxg^{-1} = x\}$.

Since $gxg^{-1} = x \iff gx = xg$, we have $G_x = C_G(x)$, where $C_G(x) = \{g \in G : gx = xg\}$ is the centralizer of x in G.

By the Orbit–Stabilizer Theorem,

$$|\operatorname{Cl}(x)| = [G : G_x] = [G : C_G(x)]. \tag{17}$$

Now, $x \in Z(G) \iff gx = xg$ for all $g \in G$, and therefore $x \in Z(G) \iff C_G(x) = G \iff |\operatorname{Cl}(x)| = 1$.

Thus the conjugacy classes contained in $Z(G)$ are precisely the singleton conjugacy classes. If $\operatorname{Cl}(x_1), \dots, \operatorname{Cl}(x_r)$ are the distinct non-singleton conjugacy classes, then

$$G = Z(G) \cup \operatorname{Cl}(x_1) \cup \cdots \cup \operatorname{Cl}(x_r). \tag{18}$$

Taking cardinalities and applying the Orbit–Stabilizer Theorem,

$$\begin{aligned} |G| &= |Z(G)| + \sum_{i=1}^{r} |\operatorname{Cl}(x_i)| \\ &= |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)]. \end{aligned} \tag{19}$$

Hence, $|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)]$ is the **class equation** of G.

[5]The special linear group consists of all those elements of $\mathrm{GL_n}(\mathbb{R})$ whose determinant is 1.

Definition 3.91 (Automorphism Group). Let G be a group. The **automorphism group** of G, denoted $\mathrm{Aut}(G)$, is the set of all group isomorphisms from G to itself:

$$\mathrm{Aut}(G) := \{\varphi : G \to G \mid \varphi \text{ is an isomorphism}\}.$$

The group operation is given by composition of maps, i.e.,

$$(\varphi \circ \psi)(g) = \varphi(\psi(g)) \quad \text{for all } g \in G.$$

Example 3.92. Let us consider $\mathbb{R}^n$ as a vector space over $\mathbb{R}$. By definition, $\mathrm{Aut}(\mathbb{R}^n)$ consists of all bijective linear maps $T : \mathbb{R}^n \to \mathbb{R}^n$. Fix the standard basis $\{e_1, \ldots, e_n\}$ of $\mathbb{R}^n$. Then every linear transformation T is uniquely determined by its action on the basis and hence can be represented by a matrix $A \in M_n(\mathbb{R})$ such that $T(x) = Ax$ for all $x \in \mathbb{R}^n$. Moreover, T is bijective iff A is invertible, which follows iff $\det(A) \neq 0$, i.e., $A \in \mathrm{GL}_n(\mathbb{R})$.

Define a map

$$\Phi : \mathrm{GL}_n(\mathbb{R}) \to \mathrm{Aut}(\mathbb{R}^n), \quad A \mapsto (x \mapsto Ax).$$

We claim that this map is a group homomorphism, since

$$\Phi(AB)(x) = ABx = A(Bx) = \Phi(A)(\Phi(B)(x))$$

so $\Phi(AB) = \Phi(A) \circ \Phi(B)$.

The map is injective because if $\Phi(A) = \Phi(B)$, then $Ax = Bx$ for all $x \in \mathbb{R}^n$, which implies $A = B$.

The map is surjective because every automorphism of $\mathbb{R}^n$ arises from an invertible linear transformation and hence from some matrix in $\mathrm{GL}_n(\mathbb{R})$. Therefore, Φ is a group isomorphism, and hence

$$\mathrm{Aut}(\mathbb{R}^n) \cong \mathrm{GL}_n(\mathbb{R}).$$

Definition 3.93 (Inner Automorphism). Let G be a group. For each $g \in G$, define a map $\varphi_g : G \to G$ by $\varphi_g(x) = gxg^{-1}$. Such maps are called **inner automorphisms**. The set of all inner automorphisms is denoted by $\mathrm{Inn}(G)$:

$$\mathrm{Inn}(G) := \{\varphi_g \mid g \in G\}.$$

3.94. Note that each φ_g is an automorphism since it is bijective with inverse $\varphi_{g^{-1}}$, and it is also a homomorphism because $\varphi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\varphi_g(y)$.

Theorem 3.95. $\mathrm{Inn}(G)$ is a group under composition.

Proof. For $g, h \in G$, we have

$$\varphi_g \circ \varphi_h(x) = \varphi_g(hxh^{-1}) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = \varphi_{gh}(x), \quad \Rightarrow \varphi_g \circ \varphi_h = \varphi_{gh}$$

The identity element is φ_e, since $\varphi_e(x) = exe^{-1} = x$ for all $x \in G$.

The inverse of φ_g is $\varphi_{g^{-1}}$, since

$$\varphi_g \circ \varphi_{g^{-1}} = \varphi_e = \varphi_{g^{-1}} \circ \varphi_g.$$

Hence $\mathrm{Inn}(G)$ is a group under composition. □

Definition 3.96 (Semidirect Product). Let K and H be groups, and let $\varphi : H \to \mathrm{Aut}(K)$ be a group homomorphism. The *semidirect product* of K by H (with respect to φ) is the set

$$K \rtimes_\varphi H := \{(k, h) \mid k \in K,\ h \in H\},$$

with the group operation defined by

$$(k_1, h_1) \cdot (k_2, h_2) := (k_1\, \varphi(h_1)(k_2),\ h_1 h_2).$$

Example 3.97. Let G be the set of all maps

$$T_{a,b} : \mathbb{R} \to \mathbb{R}, \quad T_{a,b}(x) = ax + b,\ a \neq 0$$

then G forms a group under composition of maps.

Let A be the multiplicative group of maps

$$T_{a,0}(x) = ax,\ a \neq 0$$

then,

$$A \cong \mathbb{R}^* = \mathbb{R}\backslash\{0\}.$$

Let B be the group of all translations.

$$T_{1,b}(x) = x + b$$

Then,

$$T_{a,b} \mapsto a$$

is a homomorphism of G onto the group A. And the kernel of this map will be the group B. Since B is normal in G, we observe that

$$G = AB = BA, \quad A \cap B = \{\mathrm{id}\}.$$

G is said to be the **semidirect product** of A and B.

3.98. A semidirect product is a generalization of a direct product. Whenever φ is trivial, we obtain the following definition of a direct product.

Definition 3.99 (Direct Product). Let K and H be groups. Their direct product is defined as

$$K \times H := \{(k, h) \mid k \in K,\ h \in H\},$$

with the group operation given by

$$(k_1, h_1)(k_2, h_2) = (k_1 k_2,\ h_1 h_2).$$

EXAMPLE 3.100. Consider the direct product of $\mathbb{Z}/2\mathbb{Z}$ with itself.

$G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ consists of the following four elements:

$$G = \{(0,0), (1,0), (0,1), (1,1)\}$$

where the group operation is component-wise addition modulo 2.

$(0,0)$ is the identity element in G, and every non-identity element in has order 2.

3.101. $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is not isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

To determine if two groups of the same order are isomorphic, we examine the orders of their elements:

- In $\mathbb{Z}/4\mathbb{Z}$, there exists an element of order 4 (specifically, 1 and 3). Since there is an element that generates the entire group, $\mathbb{Z}/4\mathbb{Z}$ is cyclic.
- In $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, we calculate the order of each non-identity element:

$$(1,0) + (1,0) = (0,0)$$
$$(0,1) + (0,1) = (0,0)$$
$$(1,1) + (1,1) = (0,0)$$

Because there is no element of order 4, it cannot be isomorphic to $\mathbb{Z}/4\mathbb{Z}$. This group is known as the **Klein four-group**, briefly discussed in 3.23, denoted K_4 or V_4.

EXAMPLE 3.102. Let us consider the semidirect product of $N = \mathbb{Z}/2\mathbb{Z}$ and $H = \mathbb{Z}/2\mathbb{Z}$.

The group $\mathbb{Z}/2\mathbb{Z} = \{0,1\}$ is a cyclic group of order 2. Any automorphism must map the generator 1 to another generator. Since 1 is the only generator, the only possible automorphism is the identity map:

$$\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z}) \cong \{id\}$$

Because the automorphism group is trivial, the only possible homomorphism $\phi : \mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/2\mathbb{Z})$ is the trivial homomorphism (mapping every element to the identity).

When the action ϕ is trivial, the semidirect product is exactly the direct product:

$$\mathbb{Z}/2\mathbb{Z} \rtimes_{\mathrm{triv}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \cong K_4$$

(26 Apr, 2026) This section on groups will be updated as and when required.

3. Rings

DEFINITION 3.103 (Ring). A **ring** is a set R equipped with two binary operations, $+$ and $\cdot$, such that:

(1) $(R, +)$ is an abelian group.
(2) $(R, \cdot)$ is a semigroup.
(3) $a \cdot (b + c) = a \cdot b + a \cdot c$, $(a + b) \cdot c = a \cdot c + b \cdot c$, i.e., left and right distributivity holds.

EXAMPLE 3.104. The set of integers ($\mathbb{Z}$), rationals ($\mathbb{Q}$), reals ($\mathbb{R}$), and complex numbers ($\mathbb{C}$) form a ring with respect to usual addition and multiplication.

DEFINITION 3.105 (Unital Ring). A ring R is said to be **unital** if $(R, \cdot)$ forms a monoid, i.e., R has one unique identity element such that $1x = x$ for every $x \in R$.

3.106. Given a ring R, with additive and multiplicative identities 0 and 1, respectively. Then for all $a, b \in R$,

(1) $0 \cdot a = a \cdot 0 = 0$;
(2) $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$;
(3) $(-a) \cdot (-b) = a \cdot b$;
(4) $(na) \cdot b = a \cdot (nb) = n(a \cdot b)$ for all $n \in \mathbb{Z}$.

In the above, $na = a + \cdots + a$, where there are n copies of a in the sum.

DEFINITION 3.107 (Commutative Ring). A ring R is said to be a **commutative** ring if for $x, y \in R$, we have $x \cdot y = y \cdot x$.

EXAMPLE 3.108 (Zero Ring). A **zero ring** or trivial ring, is a unique ring (up to isomorphism) which consist of a single element, namely 0. The multiplicative and additive identites int his ring coincide.

Trivially, the zero ring is a commutative ring, where $0 \cdot 0 = 0$ and $0 + 0 = 0$.

DEFINITION 3.109 (Ring homomorphism). A **ring homomorphism** is a structure preserving map defined as

$$f\colon A \to B \tag{20}$$

where A and B are rings such that

$$f(ab) = f(a)f(b), \quad f(a + b) = f(a) + f(b), \quad \text{and } f(1) = 1.$$

Therefore also $f(a - b) = f(a) - f(b)$.

EXAMPLE 3.110. The most trivial example of homomorphism of rings is identity homomorphism $1_R\colon R \to R$ which is given by $1_R(r) = r$ for all $r \in R$.

Another example is mapping $f\colon \mathbb{Z} \to \mathbb{Q}$ by the map $f(n) = n$.

3.111. For a ring homomorphism $f\colon A \to B$ and another ring homomorphism $g\colon B \to A$, then there exist $f \circ g = 1_S$ where 1_S is **identity ring homomorphism** for ring S and $g \circ f = 1_R$ where 1_R is identity ring homomorphism for ring R, then f is called an **isomorphism**. The rings (R and S) are *isomorphic* and written $R \cong S$.

3.112. For two ring homomorphism f, g, there is another ring homomorphism h given by the composition $h = f \circ g$.

3.113. A ring isomorphism from R to itself is called **automorphism**.

Definition 3.114 (Kernel). A kernel (ker) of a ring homomorphism for $f\colon A \to B$ is defined as,

$$kerf\colon = \{r \in A | f(a) = 0\} \tag{21}$$

For two kernels $ker\ f$ and $ker\ g$ we have $ker\ f + ker\ g$ is also a kernel, as well as their composition.

Definition 3.115 (Subring). Given a ring homomorphism defined by $f\colon A \to B$ which is an inclusion of the sets and the operations are the same in A and B, then A is the **subring** of B.

Example 3.116. Consider the ring homomorphism $f : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$, where $\mathbb{Z}$ is the ring of integers and $\mathbb{Z}/3\mathbb{Z}$ is the ring of integers modulo 3. The kernel of f is the set of all integers n such that n is congruent to 0 (mod 3).
The kernel of f is the set $\{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}$. This set is closed under addition and subtraction, it's also closed under multiplication by any element in $\mathbb{Z}$.

3.117. If we have a ring homomorphism $f\colon R \to S$ as an inclusion of set with the same operations, then R is the subset of S. So R is a ring, and a subring of S. Note that the operations on R and S are the same.

Proposition 3.118. A subset R is subring in S iff the following conditions are met

(1) Operations are closed under R. Every operation on S can be performed on R.
(2) R has an identity element $\mathbb{1}$.
(3) R is closed under additive inverse, so for every element r there exist a $\tilde{r}$ which is the additive inverse..

3.119. [Intersection of Rings] Any intersection of the collections of subrings of S is actually a subring.

If we have subrings for a ring S given which meet the conditions (1 to 3), then intersection of such rings are written as

$$R_I = R_1 \cap R_2 \cap R_3 \cap \cdots \cap R_n \tag{22}$$

3.120. [Subring Test] To check if R_I is the subring of the ring S, we check if the conditions of closure, identity and additive inverse are met. Since $a, b \in S$, a, b is in each of the subrings $R_1, R_2, R_3, \cdots, R_n$ as well, so R_I is closed under operations.

Since each of the subrings has identity and additive inverse, R_I has the identity element and additive inverse for the elements as well. Thus operations for S are also applicable to R_I. So R_I is a subring of S.

DEFINITION 3.121 (Ideal). An (two-sided) **ideal** $\mathfrak{I}$ is a subgroup of the additive group $(R, +)$ which is closed under multiplication for the elements of R.

Equivalently, for any $a, b \in \mathfrak{I}$:

$$a - b \in \mathfrak{I}$$

and for any element $r \in R$ and any element $x \in \mathfrak{I}$:

$$rx \in \mathfrak{I} \quad \text{and} \quad xr \in \mathfrak{I}$$

3.122. A **left ideal** satisfies the additive subgroup condition and only requires $rx \in \mathfrak{I}$ for all $r \in R$ and $x \in \mathfrak{I}$.

Dually, a **right ideal** satisfies the additive subgroup condition and only requires $xr \in \mathfrak{I}$ for all $r \in R$ and $x \in \mathfrak{I}$.

3.123. If the ring R is commutative (where $ab = ba$ for all elements), then left ideals, right ideals, and two-sided ideals are all exactly the same thing.

3.124. An ideal may or may not contain an identity element; if it does, then the ideal is called **proper ideal**, otherwise called **trivial ideal**.

EXAMPLE 3.125. In the ring of integers $\mathbb{Z}$, every ideal is of the form $n\mathbb{Z}$, consisting of all multiples of an integer n. While $n = 1$ yields the trivial ideal of the entire ring, values greater than 1, i.e. $n \geq 2$ generate non-trivial ideals, such as the set of even numbers.

3.126. We can generalize this notion to a normal subgroup one. Since ideals help us to write the quotient rings.

DEFINITION 3.127 (Quotient Rings). For a ring R, a quotient ring (given by ideal) is $R/\mathfrak{I}$ and there exist a ring homomorphism between R and $R/\mathfrak{I}$

$$f\colon R \to R/\mathfrak{I} \tag{23}$$

and the kernel of this homomorphism is $R/\mathfrak{I}$. The image of this homomorphism is given by a subring in $R/\mathfrak{I}$.

EXAMPLE 3.128. Let $\mathbb{Z}$ be the ring of integers. The quotient ring $\mathbb{Z}/2\mathbb{Z}$ is formed by identifying two integers a and b as equivalent if they have the same equivalence class (or coset).

The set of equivalence classes forms the underlying set of the quotient ring $\mathbb{Z}/2\mathbb{Z}$, and the addition and multiplication in the quotient ring are defined by:

$$(0 + 2\mathbb{Z}) + (1 + 2\mathbb{Z}) = 1 \tag{24}$$

$$(0 + 2\mathbb{Z}) * (1 + 2\mathbb{Z}) = 0 \tag{25}$$

In this example, the ring $\mathbb{Z}/2\mathbb{Z}$ has only two elements, the elements 0 and 1, and the addition and multiplication are equivalent to the operations modulo 2.

We discuss some types of elements which occur in rings.

3.129. Let a be an element of a ring R, we say that a is a **unit** if a has a multiplicative inverse, i.e., there exists an element $b \in R$ such taht $ab = ba = 1$. We also refer to a as an **invertible** element of R and b to be the **inverse** of a and vice versa. Similarly, b is also a unit.

In fact, it is also possible that a is it's own inverse, in which case $a = b$. The set of all elements of R which are units form a **group of units**, denoted as $R^{\star}$.

3.130. Let a be an element of a ring R, we say that a is a **zero divisor** if $a \neq 0$ and there is a nonzero element $b \in R$ such that $ab = ba = 0$.

3.131. Let a be an elemnet of a ring R, we say that a is a **nilpotent** if for some $k \in \mathbb{N}$, $a^k = 0$. k is then referred to as the degree of nilpotentcy.

Some mathematicians consider an element a to be nilpotent for $k = 2$, i.e., $a^2 = 0$, where a is non-zero essentailly states that a is a nilpotent element.

3.132. Let a be an elemnet of a ring R, we say that a is **idempotent** if $a^2 = a$.

A **boolean ring** is the one which consists of only idempotents.

3.133. Generally, a nilpotent element is also a zero-divisor (unless ring is a zero ring), however, the converse of the statement is not necessarily true.

Definition 3.134 (Integral Domain). An **integral domain** is a ring if it has no zero-divisors and is commutative.

Example 3.135. The ring $\mathbb{Z}$ is an integral domain, since it is commutative and has no zero-divisors. Indeed, if $ab = 0$ for $a, b \in \mathbb{Z}$, then either $a = 0$ or $b = 0$.

Example 3.136. The ring $\mathbb{Z}_6$ is not an integral domain, since it contains zero-divisors. For instance, $2 \cdot 3 = 0 \pmod 6$, even though neither 2 nor 3 is zero in $\mathbb{Z}_6$.

Definition 3.137 (Ring Extension). Let R and S be rings. We say that S is a *ring extension* of R if there exists an injective ring homomorphism

$$\varphi : R \hookrightarrow S.$$

Identifying R with its image $\varphi(R)$, we may regard R as a subring of S, and write $R \subseteq S$. Here, S is called an extension ring of R, where we denote the ring extension notation as S/R.

If $A \subseteq S$, the smallest subring of S containing both R and A is denoted $R[A]$ and is called the ring obtained from R by adjoining the elements of A.

Example 3.138. The polynomial ring $\mathbb{Z}[x]$ is a ring extension of $\mathbb{Z}$, where $\mathbb{Z} \subseteq \mathbb{Z}[x]$. It is obtained by adjoining the indeterminate x to $\mathbb{Z}$.

EXAMPLE 3.139. The ring

$$\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$$

is a ring extension of $\mathbb{Z}$. It may also be expressed as

$$\mathbb{Z}[\sqrt{2}] \cong \mathbb{Z}[x]/(x^2 - 2),$$

which corresponds to adjoining a root of the polynomial $(x^2 - 2)$.

EXAMPLE 3.140. The field extension $\mathbb{Q} \subseteq \mathbb{R}$ is a ring extension, since $\mathbb{Q}$ may be viewed naturally as a subring of $\mathbb{R}$.

DEFINITION 3.141 (Prime Ideal). Let R be a commutative ring. A proper ideal $\mathfrak{p} \subset R$ is called a **prime ideal** if it satisfies the following condition: for any elements $a, b \in R$, if the product $ab \in \mathfrak{p}$, then either $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.

Equivalently, a proper ideal $\mathfrak{p}$ is prime iff the quotient ring $R/\mathfrak{p}$ is an integral domain.

EXAMPLE 3.142. The ideal generated by a prime number p, denoted (p) or $p\mathbb{Z}$, is a prime ideal. Consider the ideal $3\mathbb{Z}$. If $ab \in 3\mathbb{Z}$, then 3 divides ab. By Euclid's lemma, 3 must divide a or 3 must divide b, meaning $a \in 3\mathbb{Z}$ or $b \in 3\mathbb{Z}$.

DEFINITION 3.143 (Zero Ideal). Let R be an integral domain then the zero ideal (0) is a prime ideal. This follows directly from the definition of an integral domain: $ab \in (0) \implies ab = 0$, which implies $a = 0$ or $b = 0$.

EXAMPLE 3.144. Let k be a field and consider the polynomial ring $k[x, y]$. The ideal generated by the indeterminate x, denoted (x), is a prime ideal.

Consider the quotient ring: $k[x, y]/(x) \cong k[y]$. Since $k[y]$ is a polynomial ring over a field, it is an integral domain, and hence (x) prime.

3.145. The ideal $(6) = 6\mathbb{Z}$ is *not* a prime ideal. Since $2 \cdot 3 = 6 \in (6)$, but neither $2 \in (6)$ nor $3 \in (6)$. Consequently, the quotient ring $\mathbb{Z}/6\mathbb{Z}$ is not an integral domain because it contains zero-divisors $(\bar{2} \cdot \bar{3} = \bar{0})$.

3.146. In $\mathbb{R}[x]$, the ideal $(x^2 - 1)$ is not prime because $(x - 1)(x + 1) = x^2 - 1 \in (x^2 - 1)$, but neither $(x - 1)$ nor $(x + 1)$ belongs to the ideal $(x^2 - 1)$.

Now that we have discussed polynomial rings and quotient rings, we can look into the following example.

EXAMPLE 3.147. An example of ring homomorphism in a ring of polynomials $R[x]$ is as follows.Let $R[x]$ be the polynomial ring, and let $R[x]/(x^2 + 1)$ be the quotient ring of polynomials modulo $x^2 + 1$. We can define a ring homomorphism $f \colon R[x] \to R[x]/(x^2 + 1)$ as follows:

$$f(p(x)) = p(x) + (x^2 + 1) \tag{26}$$

It is easy to verify that this function preserves the ring operations. For example, for any polynomials $p(x), q(x)$ in $R[x]$, we have:

$$f(p(x) + q(x)) = (p(x) + q(x)) + (x^2 + 1) = f(p(x)) + f(q(x)) \tag{27}$$

and

$$f(p(x) * q(x)) = (p(x) * q(x)) + (x^2 + 1) = f(p(x)) * f(q(x)) \tag{28}$$

This function f is an example of a ring homomorphism between $R[x]$ and its quotient group $R[x]/(x^2 + 1)$.

DEFINITION 3.148 (Maximal Ideal). Let R be a commutative ring with unity. A proper ideal $\mathfrak{m} \subset R$ is called a **maximal ideal** if there is no proper ideal I of R such that $\mathfrak{m} \subsetneq I \subsetneq R$.

Equivalently, $\mathfrak{m}$ is a maximal ideal iff the quotient ring $R/\mathfrak{m}$ is a field.

EXAMPLE 3.149. In the ring of integers $\mathbb{Z}$, the ideal generated by a prime number p, denoted $(p) = p\mathbb{Z}$, is a maximal ideal. The quotient ring $\mathbb{Z}/p\mathbb{Z}$ is isomorphic to the finite field $\mathbb{F}_p$.

EXAMPLE 3.150. In the polynomial ring $\mathbb{R}[x]$, the ideal generated by x, denoted (x), is maximal because the quotient $\mathbb{R}[x]/(x) \cong \mathbb{R}$, which is a field.

THEOREM 3.151. Let R be a commutative ring with unity, then every maximal ideal of R is a prime ideal.

PROOF. Let $\mathfrak{m}$ be a maximal ideal of R. By definition, the quotient ring $R/\mathfrak{m}$ is a field. Since every field is an integral domain, $R/\mathfrak{m}$ is an integral domain. Therefore, $\mathfrak{m}$ must be a prime ideal. The converse is generally false. □

DEFINITION 3.152 (Principal Ideal Domain (PID)). An integral domain R is called a **Principal Ideal Domain (PID)** if every ideal in R is a principal ideal. That is, for any ideal $\mathfrak{I} \subseteq R$, there exists an element $a \in R$ such that $\mathfrak{I} = (a) = \{ra \mid r \in R\}$.

EXAMPLE 3.153. The ring of integers $\mathbb{Z}$ is a PID because every ideal is of the form $n\mathbb{Z} = (n)$ for some integer n.

EXAMPLE 3.154. In the polynomial ring $\mathbb{Z}[x]$, the ideal (x) is prime (since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, an integral domain), but it is **not** maximal because $\mathbb{Z}$ is not a field. Notice that (x) is strictly contained in the maximal ideal $(2, x)$. This shows that $\mathbb{Z}[x]$ is not a PID.

DEFINITION 3.155 (Unique Factorization Domain (UFD)). An integral domain R is called a **Unique Factorization Domain (UFD)** if every non-zero, non-unit element $r \in R$ satisfies two conditions:

(1) r can be written as a product of irreducible elements: $r = p_1 p_2 \cdots p_n$.

(2) This factorization is unique up to the order of the factors and multiplication by units. That is, if $r = q_1 q_2 \cdots q_m$ is another factorization into irreducibles, then $n = m$, and after possibly reordering, each p_i is an associate of q_i (meaning $p_i = uq_i$ for some unit u).

EXAMPLE 3.156. The ring $\mathbb{Z}$ is a UFD[6]. The polynomial ring $\mathbb{Z}[x]$ is also a UFD, even though it is not a PID.

3.157. An example of an integral domain that is **not** a UFD is $\mathbb{Z}[\sqrt{-5}]$. In this ring, the number 6 has two distinct irreducible factorizations:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

DEFINITION 3.158 (Multiplicative Set). Let R be a commutative ring with unity. A subset $S \subseteq R$ is called a **multiplicative set** if the following conditions hold:

(1) $1 \in S$,
(2) $0 \notin S$,
(3) for all $a, b \in S$, we have $ab \in S$.

That is, S is closed under multiplication.

DEFINITION 3.159 (Localization). Let R be a commutative ring with unity and let $S \subseteq R$ be a multiplicative set. The **localization** of R at S is the ring $S^{-1}R$ consisting of equivalence classes of formal fractions

$$\frac{r}{s}, \quad r \in R,\ s \in S, \quad \text{where} \quad \frac{r}{s} \sim \frac{r'}{s'}$$

if there exists $u \in S$ such that

$$u(rs' - r's) = 0.$$

Addition and multiplication are defined by

$$\frac{r}{s} + \frac{r'}{s'} = \frac{rs' + r's}{ss'},$$

and

$$\frac{r}{s} \cdot \frac{r'}{s'} = \frac{rr'}{ss'}.$$

There is a natural ring homomorphism

$$\varphi : R \to S^{-1}R \quad \text{given by} \quad \varphi(r) = \frac{r}{1}.$$

EXAMPLE 3.160. Let $S = \mathbb{Z} \setminus \{0\}$, then localizing $\mathbb{Z}$ at all nonzero integers produces the field of rational numbers. That is, $S^{-1}\mathbb{Z} = \left\{ \frac{a}{b} : a \in \mathbb{Z},\, b \in S \right\} = \mathbb{Q}$.

[6] This is simply the Fundamental Theorem of Arithmetic.

3.161. Fix a prime number p and let $S = \mathbb{Z} \setminus (p)$. The localization $\mathbb{Z}_{(p)} := S^{-1}\mathbb{Z}$ consists of fractions $\frac{a}{b}$ where b is not divisible by p.

EXAMPLE 3.162. Consider $\mathbb{Z}_{(5)}$ then the element $\frac{7}{3}$ exists since $3 \notin (5)$, whereas $tfrac1/5$ is not an element of $\mathbb{Z}_{(5)}$.

3.163. For another example, let k be a field and consider the polynomial ring $k[x]$. Set $S = \{1, x, x^2, x^3, \dots\}$. Then $S^{-1}k[x] = k[x, x^{-1}]$, the ring of Laurent polynomials. Thus localization has the effect of making the element x invertible.

3.164. The localization perspective was developed in the early twentieth century through the work of mathematicians such as Wolfgang Krull, Emmy Noether, and later Oscar Zariski. The resulting notion of a *local ring* became one of the fundamental objects of modern commutative algebra and algebraic geometry.

DEFINITION 3.165 (Local Ring). A commutative ring R with identity is called a *local ring* if it possesses a unique maximal ideal.

THEOREM 3.166. *Let R be a commutative ring with identity. The following are equivalent:*

(1) *R is a local ring.*
(2) *The set of non-units of R forms an ideal.*

3.167. The ring $\mathbb{Z}_{(p)}$ is a local ring. Its unique maximal ideal is $(p)\mathbb{Z}_{(p)}$. Since every element not divisible by p is a unit.

Also, every field k is a local ring. Its unique maximal ideal is (0).

EXAMPLE 3.168. The ring $k[x]_{(x)}$ is a local ring whose unique maximal ideal is $(x)k[x]_{(x)}$.

3.169. The ring $\mathbb{Z}$ is not local. Since for every prime number p, the ideal (p) is maximal, $\mathbb{Z}$ has infinitely many maximal ideals.

THEOREM 3.170. *Let A be a commutative ring and let $\mathfrak{p}$ be a prime ideal of A. Then the localization $A_{\mathfrak{p}}$ is a local ring whose unique maximal ideal is $\mathfrak{p}A_{\mathfrak{p}}$.*

DEFINITION 3.171 (Radical of an Ideal). Let R be a commutative ring and let $\mathcal{I} \subseteq R$ be an ideal. The *radical* of $\mathcal{I}$, denoted by $\sqrt{\mathcal{I}}$, is defined as

$$\sqrt{\mathcal{I}} = \{\, a \in R \mid a^n \in \mathcal{I} \text{ for some } n \geq 1 \,\}.$$

An ideal $\mathcal{I}$ is called a *radical ideal* if

$$\mathcal{I} = \sqrt{\mathcal{I}}.$$

Equivalently, an ideal $\mathcal{I}$ is radical if whenever $a^n \in \mathcal{I}$ for some positive integer n, then $a \in \mathcal{I}$.

3.172. Every prime ideal is a radical ideal.

Let $\mathfrak{p}$ be a prime ideal and suppose $a^n \in \mathfrak{p}$ for some $n \geq 1$. Since $a^n = a \cdot a^{n-1}$, the defining property of prime ideals implies that $a \in \mathfrak{p}$. Thus, $\sqrt{\mathfrak{p}} = \mathfrak{p}$. The converse is not true in general, i.e., a radical ideal need not be prime.

Example 3.173. Consider the ideal $\mathcal{I} = (x^2) \subseteq k[x]$. Since $x^2 \in \mathcal{I}$, we have $x \in \sqrt{\mathcal{I}}$. More generally, if $f(x)^n \in (x^2)$ for some $n \geq 1$, then x divides $f(x)$, so $f(x) \in (x)$. Hence $\sqrt{(x^2)} = (x)$. Therefore (x^2) is not a radical ideal, whereas (x) is.

Definition 3.174 (Nilradical). Let R be a commutative ring. The *nilradical* of R, denoted by $\mathrm{Nil}(R)$, is the set of all nilpotent elements of R, that is,

$$\mathrm{Nil}(R) = \{\, a \in R \mid a^n = 0 \text{ for some } n \geq 1 \,\}.$$

Equivalently, $\mathrm{Nil}(R) = \sqrt{(0)}$, where (0) denotes the zero ideal of R.

Proposition 3.175. The nilradical of a commutative ring is a radical ideal.

Proof. Since $\mathrm{Nil}(R) = \sqrt{(0)}$, it suffices to prove that the radical of any ideal is an ideal.

Let $a, b \in \sqrt{(0)}$. Then there exist positive integers m, n such that $a^m = 0$, and $b^n = 0$.

Using the binomial theorem,

$$(a+b)^{m+n-1} = \sum_{i=0}^{m+n-1} \binom{m+n-1}{i} a^i b^{m+n-1-i}.$$

Each summand contains either a^m or b^n and is therefore 0. Hence $(a+b)^{m+n-1} = 0$, so $a + b$ is nilpotent.

If $r \in R$, then $(ra)^m = r^m a^m = 0$, since R is commutative.

Therefore ra is nilpotent.

Hence $\mathrm{Nil}(R)$ is an ideal.

And, since $\mathrm{Nil}(R) = \sqrt{(0)}$, it follows immediately that it is radical. □

Definition 3.176 (Reduced Ring). A commutative ring R is called *reduced* if it contains no nonzero nilpotent elements.

Equivalently, $\mathrm{Nil}(R) = 0$.

Definition 3.177 (Spectrum). Let R be a commutative ring. The *spectrum* of R, denoted by $\mathrm{Spec}(R)$, is the set of all prime ideals of R, that is,

$$\mathrm{Spec}(R) = \{\mathfrak{p} \subseteq R \mid \mathfrak{p} \text{ is a prime ideal}\}.$$

3.178. We enter into a discussion of locally reduced rings henceforth, and as the name suggests, we focus on reducedness being a local property, whereas the underlying philosophy is the recovery of global properties via local ones. This discuission will make more sense when we delve into ringed spaces and schemes.

DEFINITION 3.179 (Locally Reduced Ring (1)). A commutative ring R is said to be *locally reduced* if, for every prime ideal $\mathfrak{p}$ of R, the localization $R_{\mathfrak{p}}$ is a reduced ring.

3.180. Essentially, we define a locally reduced ring based ont he collection of all prime ideals, which we now know to be, is $\mathrm{Spec}(R)$, and hence the above defintion can be rephrased as: A commutative ring R is said to be *locally reduced* if, for every prime ideal $\mathfrak{p} \in \mathrm{Spec}(R)$, the localization $R_{\mathfrak{p}}$ is a reduced ring.

DEFINITION 3.181 (Locally Reduced Ring (2)). A commutative ring R is said to be *locally reduced* if, for every maximal ideal $\mathfrak{m}$ of R, the localization $R_{\mathfrak{m}}$ is a reduced ring.

EXAMPLE 3.182. Let k be a field. Consider the polynomial ring $k[x]$ and let $\mathfrak{m} = (x)$. The localization $R = k[x]_{\mathfrak{m}}$ is a local ring whose unique maximal ideal is

$$\mathfrak{m}R = (x)k[x]_{\mathfrak{m}}.$$

Since $k[x]$ is an integral domain and localization preserves integral domains, R is an integral domain. Consequently, R is reduced.

EXAMPLE 3.183 (The ring of integers). The ring of integers $\mathbb{Z}$ is a reduced ring. Suppose that $a \in \mathbb{Z}$ is nilpotent. Then there exists a positive integer n such that $a^n = 0$.

Since $\mathbb{Z}$ is an integral domain, it contains no zero divisors. Consequently, $a^n = 0 \implies a = 0$.

Hence $\mathbb{Z}$ contains no nonzero nilpotent elements, and therefore $\mathbb{Z}$ is reduced.

3.184. $\mathbb{Z}$ is not a local ring, since it possesses more than one maximal ideal[7] Thus $\mathbb{Z}$ provides a simple example of a reduced ring that is not local.

EXAMPLE 3.185 (A reduced ring). Now, let k be a field and consider the localization $R = k[x]_{(x)}$.

Since (x) is the unique maximal ideal of R, the ring R is local. Moreover, $k[x]$ is an integral domain, and localization preserves integral domains. Hence R is also an integral domain, and therefore reduced.

EXAMPLE 3.186 (A locally reduced ring). Since $\mathbb{Z}$ is reduced, every localization $\mathbb{Z}_{\mathfrak{p}}$, where $\mathfrak{p}$ is a prime ideal of $\mathbb{Z}$, is also reduced. Consequently, $\mathbb{Z}$ is locally reduced.

More generally, every reduced ring is locally reduced. In fact, for commutative rings, the converse also holds, so that a ring is reduced if and only if it is locally reduced.

[7]For every prime number p, the principal ideal (p) is maximal in $\mathbb{Z}$.

Definition 3.187 (Submodules). Let M be an $R-$module. A subset $N \subseteq M$ is called a *submodule* of M if N itself is an $R-$module under the addition and scalar multiplication inherited from M.

3.188. Equivalently, $N \subseteq M$ is a submodule if

$$0 \in N, \quad x + y \in N \quad \text{for all } x, y \in N, \, rx \in N \quad \text{for all } r \in R, \; x \in N.$$

Example 3.189. Consider $M = \mathbb{Z}$ as a $\mathbb{Z}$-module. For any $n \in \mathbb{Z}$,

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is a submodule of $\mathbb{Z}$. Indeed, if $na, nb \in n\mathbb{Z}$, then

$$na + nb = n(a + b) \in n\mathbb{Z},$$

and for $r \in \mathbb{Z}$,

$$r(na) = n(ra) \in n\mathbb{Z}.$$

3.190. Thus, every ideal of a commutative ring (R) can be viewed naturally as a submodule of the $R-$module R itself.

Definition 3.191 (Quotient Modules). Let M be an $R-$module and let N be a submodule of M. Define an equivalence relation on M by

$$x \sim y \quad \Longleftrightarrow \quad x - y \in N.$$

The equivalence class of $x \in M$ is called the *coset of N represented by x* and is denoted by

$$x + N = \{x + n : n \in N\}.$$

The set of all such cosets is called the *quotient module* of M by N, denoted

$$M/N = \{x + N : x \in M\}.$$

The module operations are defined by

$$(x + N) + (y + N) = (x + y) + N$$

and

$$r(x + N) = rx + N, \qquad r \in R.$$

These operations are well-defined because N is a submodule of M.

Example 3.192. Take $M = \mathbb{Z}$ and $N = 2\mathbb{Z}$. Then the quotient module consists of elements of the form

$$\mathbb{Z}/2\mathbb{Z} = 0 + 2\mathbb{Z}, 1 + 2\mathbb{Z}.$$

Thus the quotient module consists of the two residue classes of even and odd integers. More generally, for a commutative ring R and an ideal $I \subseteq R$, R/I is simultaneously a quotient ring and a quotient R-module.

Definition 3.193 (Finitely Generated Modules). Let M be an R-module. We say that M is *finitely generated* if there exist finitely many elements

$$m_1, m_2, \ldots, m_n \in M$$

such that every $m \in M$ can be expressed as

$$m = r_1 m_1 + r_2 m_2 + \cdots + r_n m_n$$

for some $r_1, r_2, \ldots, r_n \in R$. In this case, we write

$$M = Rm_1 + \cdots + Rm_n$$

and say that $m_1, \ldots, m_n$ generate M.

3.194. Equivalently, if M is a finitely generated $R-$module, we can write the following

$$M = \langle m_1, \ldots, m_n \rangle_R.$$

Example 3.195. The $\mathbb{Z}$-module $\mathbb{Z}^2$ is finitely generated. Indeed,

$$\mathbb{Z}^2 = \mathbb{Z}(1,0) + \mathbb{Z}(0,1).$$

Thus $(1,0)$ and $(0,1)$ generate $\mathbb{Z}^2$.

Example 3.196. Let $R = k[x]$. Consider $M = k[x]$ as an R-module. It is generated by the single element 1, since

$$f(x) = f(x) \cdot 1$$

for every $f(x) \in k[x]$. Hence

$$M = R \cdot 1.$$

3.197. An important distinction in the above two examples is that a module may be generated by infinitely many elements without having a finite generating set.

Definition 3.198 (Module Homomorphisms). Let M and N be R-modules. A map

$$f : M \longrightarrow N$$

is called an *R-module homomorphism* if, for all $x, y \in M$ and $r \in R$,

$$f(x+y) = f(x) + f(y)$$

and

$$f(rx) = rf(x).$$

3.199. An R-module homomorphism is a map that preserves both the additive structure and scalar multiplication.

EXAMPLE 3.200. Define

$$f : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad f(a,b) = a + b.$$

Then

$$\begin{aligned} f((a,b) + (c,d)) &= f(a+c, b+d) \\ &= a + c + b + d \\ &= f(a,b) + f(c,d), \end{aligned}$$

and for $n \in \mathbb{Z}$,

$$f(n(a,b)) = na + nb = nf(a,b).$$

Hence, f is a $\mathbb{Z}$-module homomorphism.

DEFINITION 3.201 (Kernel and Image). Let $f : M \longrightarrow N$ be an R-module homomorphism. The **kernel** of f is defined as

$$\ker f := \{x \in M \mid f(x) = 0\},$$

while the **image** of f is defined as

$$\operatorname{im} f := \{f(x) \mid x \in M\}.$$

The kernel $\ker f$ is a submodule of M, while $\operatorname{im} f$ is a submodule of N.

For the homomorphism

$$f : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad f(a,b) = a + b,$$

we have

$$\ker f = \{(a,b) \in \mathbb{Z}^2 \mid a + b = 0\} = \{(a,-a) \mid a \in \mathbb{Z}\},$$

and

$$\operatorname{im} f = \mathbb{Z}.$$

3.202. A useful characterization of injectivity is

$$f \text{ is injective} \quad \Longleftrightarrow \quad \ker f = \{0\}.$$

3.203. A ring R satisfies the **ascending chain condition** (ACC) on ideals if for every ascending chain of ideals

$$\mathcal{I}_1 \subseteq \mathcal{I}_2 \subseteq \mathcal{I}_3 \subseteq \cdots \tag{29}$$

eventually stabilizes.
That is, there exists $n \in \mathbb{N}$ such that

$$\mathcal{I}_m = \mathcal{I}_n \qquad \text{for all } m \geq n. \tag{30}$$

3.204. Dually, a ring R satisfies the **descending chain condition** (DCC) on ideals if for every descending chain of ideals

$$\mathcal{I}_1 \supseteq \mathcal{I}_2 \supseteq \mathcal{I}_3 \supseteq \cdots, \tag{31}$$

eventually stabilizes.
That is, there exists $n \in \mathbb{N}$ such that

$$\mathcal{I}_m = \mathcal{I}_n \qquad \text{for all } m \geq n. \tag{32}$$

Definition 3.205 (Noetherian Ring). A ring R is called *Noetherian* if it satisfies the ascending chain condition (ACC) on ideals.

Definition 3.206 (Artinian Ring). A ring R is called *Artinian* if it satisfies the descending chain condition (DCC) on ideals.

Theorem 3.207. For a ring R, the following are equivalent:

(1) R is Noetherian.
(2) Every ideal of R is finitely generated.
(3) Every ascending chain of ideals stabilizes.

Theorem 3.208. For a ring R, the following are equivalent:

(1) R is Artinian.
(2) Every descending chain of ideals stabilizes.
(3) Every nonempty set of ideals of R has a minimal element.

3.209. Every Artinian ring is Noetherian, although a Noetherian ring need not be Artinian.

In an Artinian ring, one can use the DCC to show that every prime ideal is maximal and that the ring has only finitely many maximal ideals. More generally, the ideal structure can be decomposed into finitely many pieces, and this ultimately gives finite length, and a module of finite length is both Noetherian and Artinian.

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